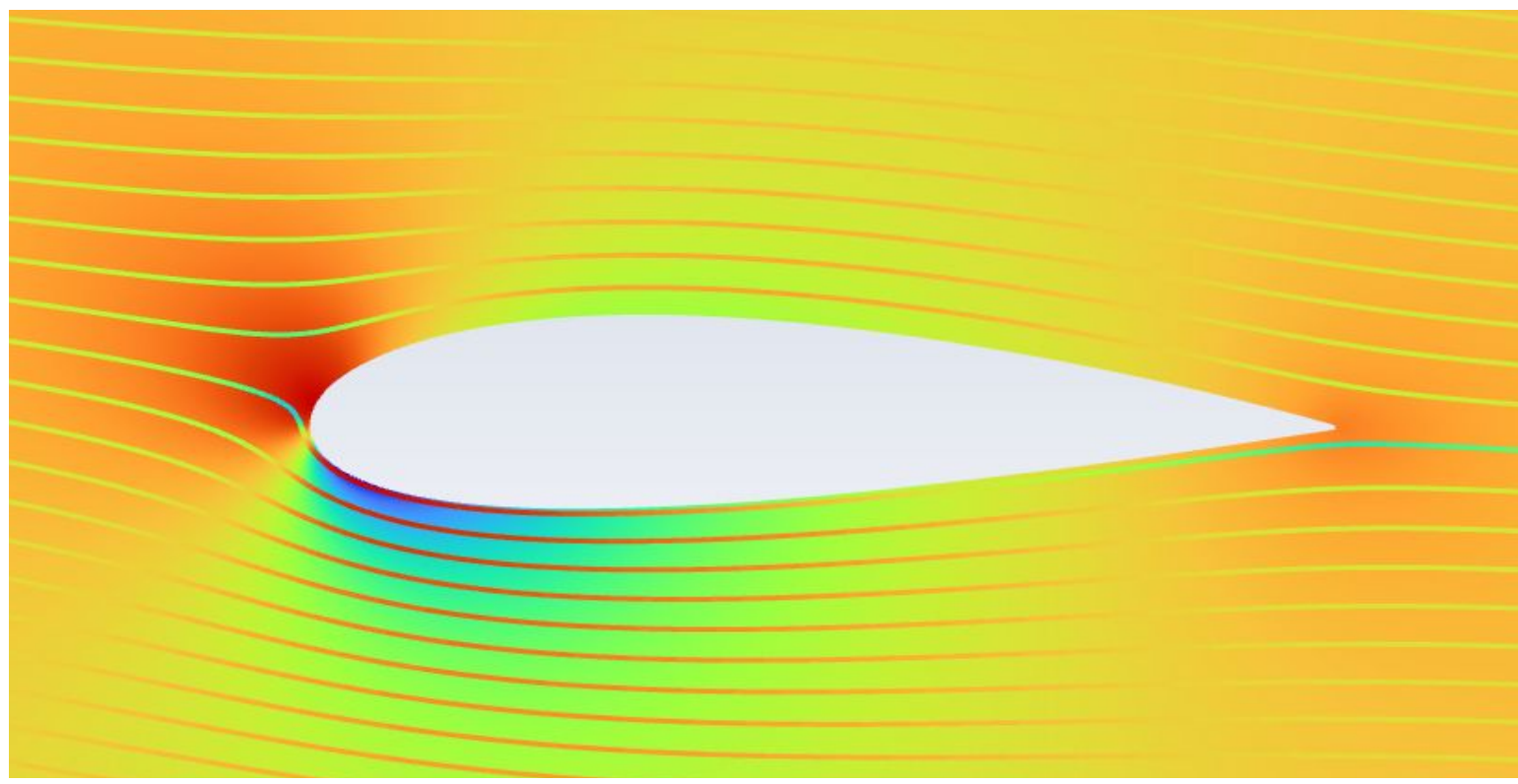




Shaping Lift: The Role of Camber and Thickness in Airfoils

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Objective of the Study

- Investigate the effect of airfoil thickness and camber
- Utilize a 2x3x2 full factorial design with blocking and replication
- Quantify main effects and identify factors that significantly influence aerodynamic performance
- Determine which parameter most strongly influences aerodynamic properties

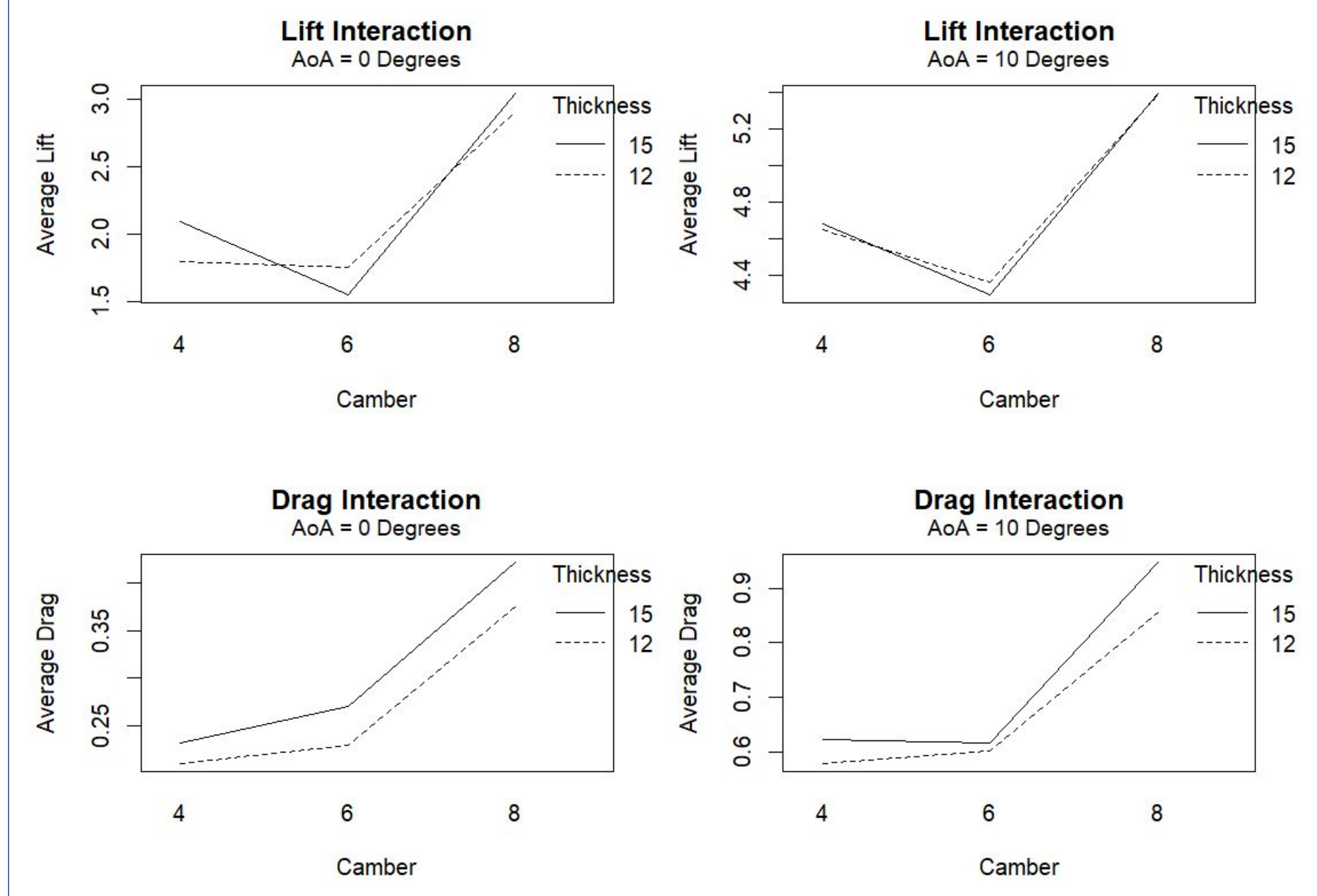
Motivation

- Performed to gain a broader understanding of aerodynamic design
- Airfoil wing design directly dictates and influences performance characteristics including lift generation and drag behavior, which vary with operating conditions such as angle of attack and flow velocity
- This relationship connects aerodynamic theory to practical design constraints and performance tradeoffs in real-world applications

Data

			Angle of Attack			
			0 Degrees		10 Degrees	
Thickness	Camber	Block	Lift [N]	Drag [N]	Lift [N]	Drag [N]
12%	4%	1, 2	1.8277, 1.738	0.2243, 0.1972	4.8161, 4.4784	0.6062, 0.5519
12%	6%	1, 2	2.0994, 1.4100	0.2604, 0.1981	4.5449, 4.1821	0.6821, 0.5221
12%	8%	1, 2	2.9315, 2.8751	0.3863, 0.3639	5.4697, 5.2999	0.8656, 0.8429
15%	4%	1, 2	2.4129, 1.7730	0.2648, 0.1981	4.9108, 4.4455	0.6944, 0.5520
15%	6%	1, 2	1.7166, 1.3949	0.3254, 0.2147	4.5261, 4.0587	0.7010, 0.5288
15%	8%	1, 2	3.3558, 2.7188	0.4864, 0.3574	5.4632, 5.3218	1.0833, 0.8122

Data Exploration

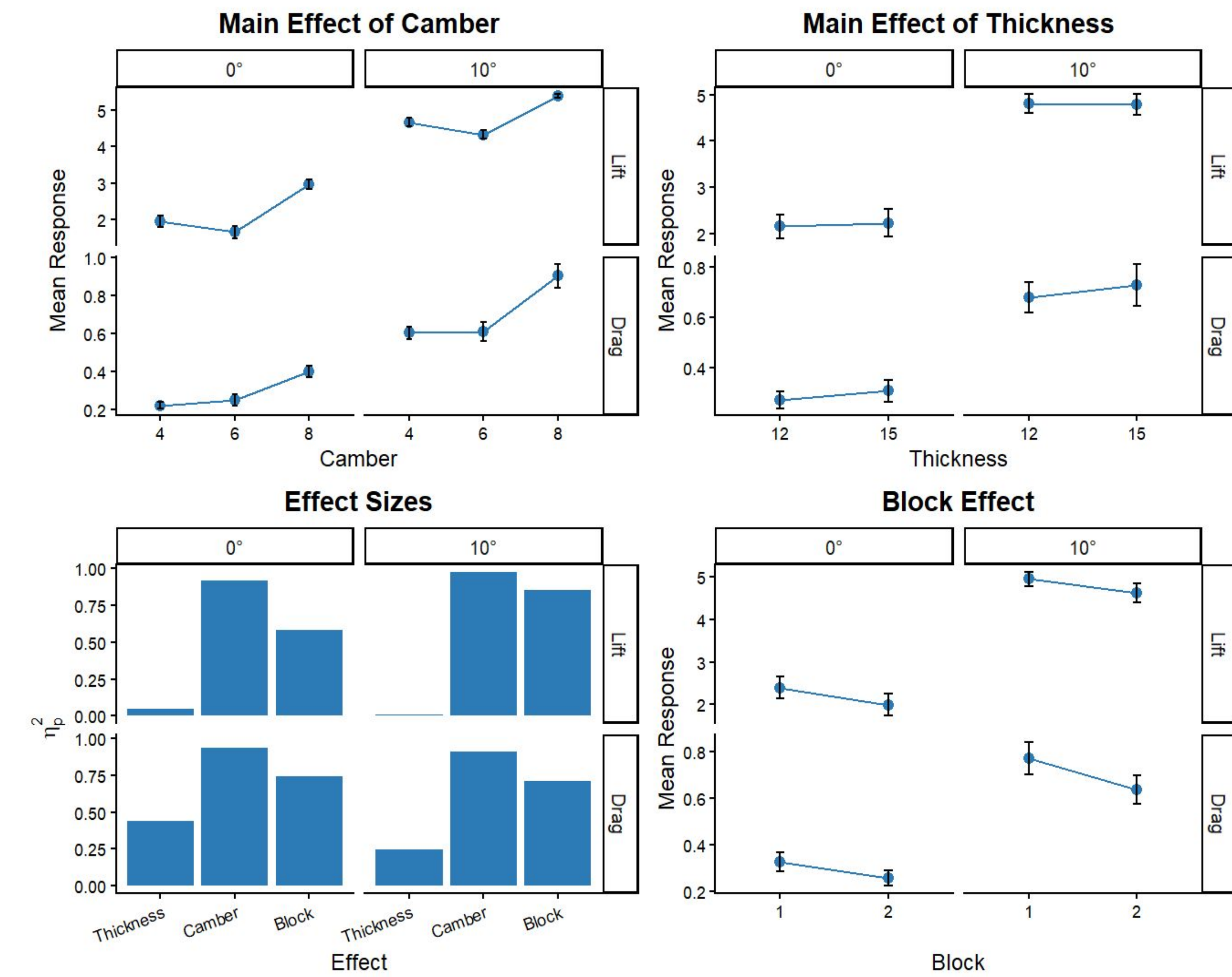


Statistical Analysis

- Model Specification: Response ~ Thickness x Camber + Block
- Interaction: Interaction effects were tested and found to be statistically insignificant in all models
- Model Refinement: Reduced model without interaction were fitted and used for final analysis
- Diagnostics: Residual plots were examined; no major violations of model assumptions were observed
- ANOVA test results that the effect of camber on both drag and lift is statistically significant at both AoA but the effect of thickness is not significant at any conditions. At 0° AoA, the thickness is close to being statistically significant

Results

- Camber showed the strongest effect on both the lift and drag force
- Thickness had smaller and less consistent effects; its influence on drag was greater than its influence on lift

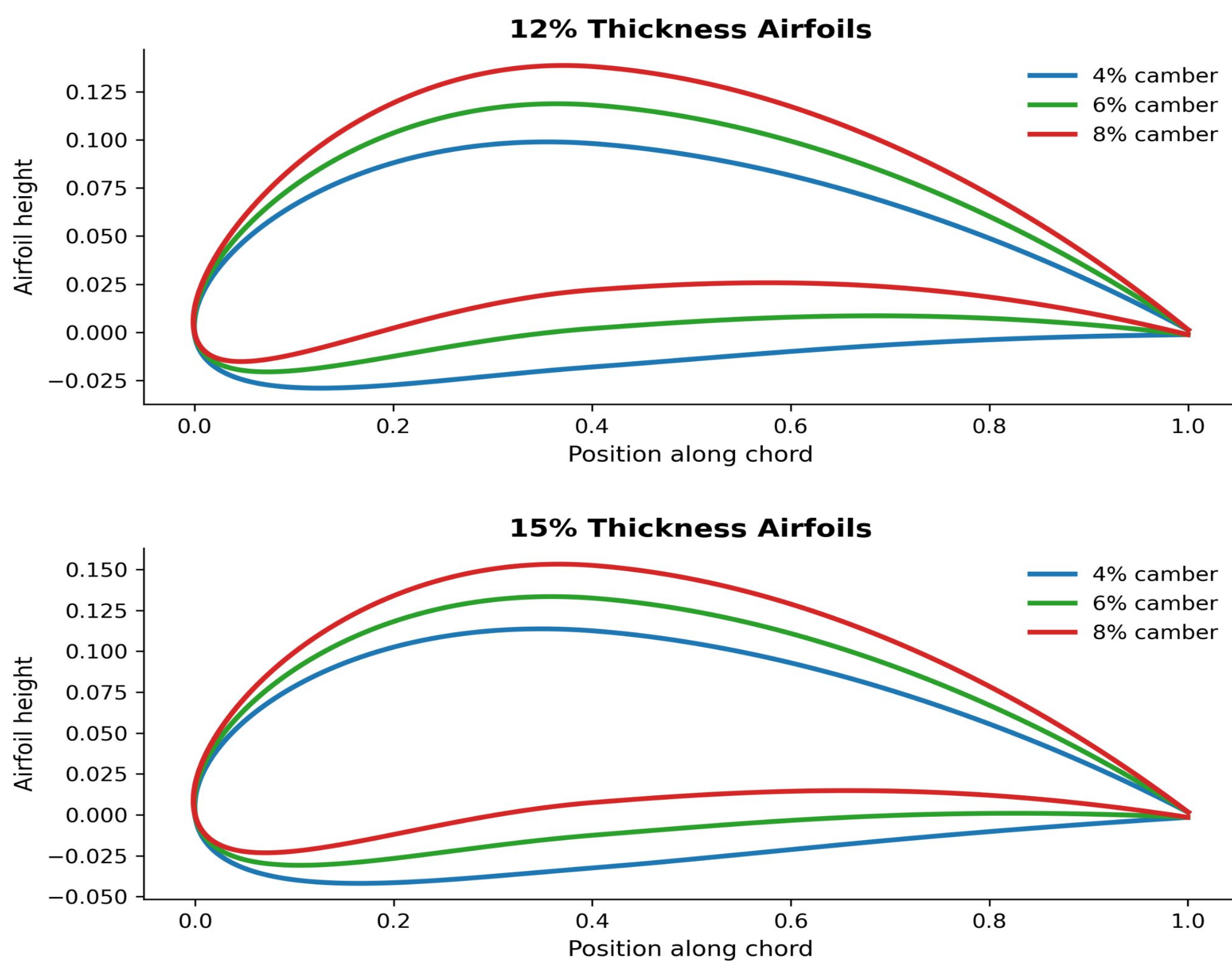
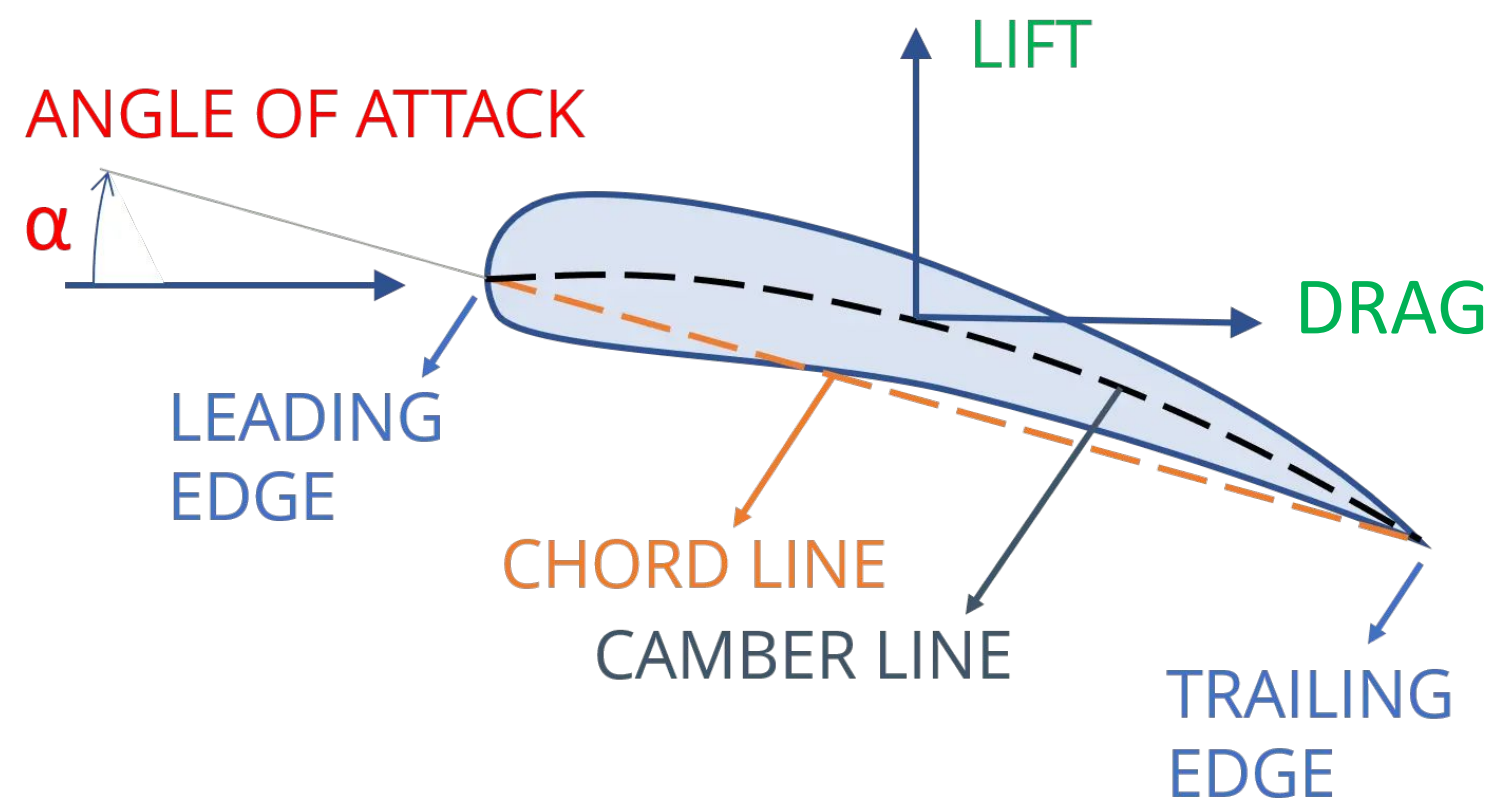


- Effect Sizes plot used the following equation to determine the proportion of total variance in the dependent variable

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{residual}}}$$

Experimental Design

- Factors:
 - Thickness: 12%, 15%
 - Camber: 4%, 6%, 8%
- These levels were selected based on aerodynamic design principles to represent a variety of airfoils with different performance characteristics
- Each airfoil was tested in the wind tunnel at 0° and 10° angle of attack and 20 m/s velocity in randomized order, with replication performed on a separate day
- Replicates were performed on different days; therefore, a blocking factor was introduced to account for day-to-day variability.



Discussion & Conclusion

- Camber had a strong and statistically significant effect on both lift and drag at 0° and 10° angle of attack
- Thickness showed a smaller and less consistent effect, with a greater influence on drag than lift
- Both camber and thickness exhibited a positive relationship with lift and drag
- No significant interaction between camber and thickness was detected
- Reduced models without interaction were fitted and used for final analysis
- Block effects were significant, indicating the presence of systematic experimental variability
- Optimizing camber is more important than thickness in aero optimization

Reference

Jacobs, Eastman N., Ward, Kenneth E., and Pinkerton, Robert M. The Characteristics of 78 Related Airfoil Sections from Tests in the Variable-Density Wind Tunnel. NACA Report No. 460, 1933.

Acknowledgement

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